Xiteng Yao

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Computer Engineering PhD candidate graduating in May 2027, researching LLM training and inference systems, SmartNIC/FPGA acceleration, RDMA integration, and collective communication. Internship experience at Arm and AWS includes LLM runtime development and GPU performance profiling and modeling.

Education

Boston University — Boston, MA
PhD in Computer Engineering — Expected May 2027
GPA: 4.0/4.0

Boston University — Boston, MA
BS in Computer Engineering, Minor in Computer Science, Concentration in Machine Learning — Graduated May 2023
GPA: 3.9/4.0, Magna Cum Laude

Skills

- Coding: C, C++, Python, Verilog, SystemVerilog, CUDA, Bash, MATLAB, Java, C#, JavaScript
- Tools and platforms: vLLM, TensorRT-LLM, Arm, MIPS, RISC-V, AWS, Azure, LLVM, GCC, Linux, Docker, Kubernetes, Git, CI/CD, Xilinx Vivado, Vitis HLS, Agile/Scrum

Experience

Arm Inc. — Seattle, WA
AI Research Scientist Intern — May 2026 – Aug. 2026

- Built **one of the first Attention-FFN Disaggregation (AFD)** runtimes for LLM inference, separating attention and feed-forward execution into distinct serving stages.
- Implemented **context-length-aware request routing** and **expert routing** within the AFD control system to address workload and expert imbalance.
- Compared with vLLM, the system achieved 44–67% higher token throughput while reducing p99 inter-token latency (ITL) by up to 28%.

Amazon Web Services (AWS) — Santa Clara, CA
Applied Scientist Intern, L5 — Aug. 2025 – Dec. 2025

- Applied and quantified the effects of **LLM inference-serving optimizations** (prefill/decode disaggregation, continuous batching, prefix caching) in LLM serving deployments; tuned **vLLM** runtime parameters to improve deployment performance.
- Built **KernelSight-LM**, a **GPU-kernel-aware profiling and simulation** toolchain combining kernel-latency, communication, and host-overhead models in a discrete-event scheduler to predict end-to-end LLM inference latency and throughput. Evaluated across six model families, with **2.7% median throughput-prediction error** in the target-measured setting; supports HW/SW co-design and design-space exploration. Preprint on arXiv.

Projects

Optimizing LLM Training across Multiple GPUs — Sep. 2024 – Present

- Developed a **SmartNIC-based model** for ZeRO training, using **parameter prefetching, buffering, and dynamic scheduling** to overlap GPU/CPU computation with communication and reduce duplicate parameter gathers.
- Hardware-calibrated cycle-accurate simulation projected a **training throughput speedup of over 1.6×** for a 100B-parameter model versus GPU–CPU offload with ordinary NICs; published at ACM ICS 2025 [1].

Programmable Switches for Efficient Distributed LLM Systems — Sep. 2024 – Present

- Designed **SmartFuse-LLM**, an FPGA-based switch datapath that fuses collective communication, residual addition, and RMSNorm, eliminating a network round trip and GPU-side normalization. GPU profiling and RTL co-simulation projected **approximately 50.5% lower per-boundary latency**; published at HEART 2026 [3].
- Developed **PoolStateLens** to evaluate **switch-attached memory pools** for LLM training, combining GB200/NVLink-5 measurements, performance modeling, and **ASTRA-sim** to identify communication bottlenecks and capacity–bandwidth tradeoffs; accepted at IEEE HPEC 2026 [8].
- Extending this work toward **stateful, programmable switch architectures**, implementing FPGA forward/backward datapaths and designing intermediate-state placement and workload-aware offload policies for LLM training and inference.

FPGA-Accelerated Secure Machine Learning with MPC Sep. 2023 – Present

- Designed an **FPGA architecture** for secure ML training with **multi-party computation (MPC)**, coordinating distributed computation and protocol data exchange across participating nodes.
- Integrated **Coyote's RDMA stack** on **AMD FPGA SmartNICs** for CPU-bypass memory transfers, overlapping communication with computation. Layer-level experiments achieved **2.1× convolution-layer speedup** over existing MPC frameworks; the resulting IEEE HPEC 2025 paper received the **Outstanding Student Paper Award** [2].
- Extending this work toward **GPU–FPGA acceleration** using **GPUDirect RDMA** and advanced interconnect technologies, exploring direct data-transfer paths that reduce host-memory copies and CPU involvement to improve communication efficiency and secure-training scalability.

Machine Learning–Guided Compiler Optimization for FPGAs and GPUs Apr. 2022 – Aug. 2023; 2026 – Present

- Built **RNN, Transformer, and GNN pipelines** to analyze HLS program representations and predict LLVM optimization decisions, exploring supervised and reinforcement learning approaches.
- Demonstrated **3×–23× improvements across evaluated HLS metrics**; published at **LLVM-HPC 2022** [4].
- Extended optimization to **Triton/LLVM GPU compilation on NVIDIA Blackwell**, investigating compiler-pass ordering, LLVM transformations, and **RL-guided instruction scheduling** with numerical correctness checks and measured-runtime evaluation; work accepted at IEEE HPEC 2026 [7].

Linux Operating System Implementation Sep. 2021 – Dec. 2021

- Developed a custom multithreading system for Linux, including a thread library, a round-robin scheduler, and thread-local storage (TLS) for efficient parallel task execution.
- Designed and implemented a **custom Linux file system** within the **Virtual File System (VFS)** layer.

Selected Publications

[1] SmartNIC-GPU-CPU Heterogeneous System for Large Machine Learning Model with Software-Hardware Codesign. ACM ICS 2025.

[2] Accelerating Multi-Party Computation Using Heterogeneous Systems. IEEE HPEC 2025. Outstanding Student Paper Award.

[3] SmartFuse-LLM: Fusing Computation into Collectives for Efficient Tensor-Parallel LLM Systems. HEART 2026.

[4] Reinforcement Learning Strategies for Compiler Optimization in High Level Synthesis. LLVM-HPC 2022.

[5] Load Imbalance in HPC Applications: Improved Profiling and New Ways to Use Wasted Cycles. IEEE HPEC 2025.

[6] ACBench: A Benchmark Suite for Auditing Retry and Profiling Feedback in LLM-Driven Code Optimization. IEEE HPEC 2026.

[7] Toward RL-Guided GPU Compiler Optimization: Case Studies on Nvidia Blackwell GPUs. IEEE HPEC 2026.

[8] PoolStateLens: Evaluating the Performance of an On-Switch Memory Pool for LLM Training. IEEE HPEC 2026.

Awards & Funding

- **Boston University Presidential Scholarship** (four-year, **half-tuition**) 2019–2023
- **Red Hat Collaboratory** Student Research Project Funding 2022
- BU Undergraduate Research Opportunities Program (UROP) Funding 2021
- BU Summer Training as Research Scholars (STaRS) Housing Stipend 2021

Service

- **Reviewer** for TPDS, FGCS, TRETS, MIT URTC, FCCM, FPT, H2RC, HPEC, HEART
- Co-applicant for NSF Future Computing Research Funding (NSF 25-543)

Teaching Assistantship

Department of Electrical and Computer Engineering, Boston University 2019–2026

Courses: EC311 Digital Logic Design (2 semesters); EC413 Computer Organization (4 semesters); EC440 Operating Systems; EC528 Cloud Computing; EK103 Linear Algebra; EK307 Electrical Circuits.